

AI_reference

Midterm Project - Chatroom

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Course: CS2410 Software Studio

1. AI Tool(s) Used

- Gemini (Google): mostly for looking up how to use certain APIs
- ChatGPT (OpenAI, GPT-4): asked some questions about Firebase and React stuff

Basically I used AI when I was stuck or didnt know how some API works. I would ask it a question, look at what it gives me, then write the code myself. Also even for parts that I didnt use AI to write, I would usually paste my code into chatgpt after and ask it to help me clean it up or check if theres any bugs. so a lot of the code has been polished by AI even if the original logic was mine.

2. Scope of Usage

2.1 Chatbot Integration (src/services/chatbot.js, lines 1-48)

Prompt: "how to call gemini api from javascript? like using fetch to send a message and get the response back"

Response: it showed me the fetch() call format and the JSON body structure, like the contents array with role and parts fields. also said to put the api key in the url.

What I did: I used the general idea but changed a lot. I added the systemInstruction part so the bot would act friendly. I also made it so it takes the last 10 messages from chat history as context, that way the bot kinda remembers what you were talking about. I added error handling too because my api key actually got leaked on github and stopped working lol, so I added a check for that specific error.

2.2 Sticker Canvas (src/components/StickerCanvas.jsx, lines 1-140)

Prompt: "how to make a drawing canvas in react? need to draw with mouse, different colors and brush size"

Response: gave me a basic example with useRef on a canvas element, mousedown/mousemove/mouseup handlers, and ctx.beginPath/lineTo/stroke to draw. pretty simple example with like 3 colors only.

What I did: changed a lot of stuff. added touch event support so it works on phone too. made my own color palette (12 colors) and 5 different brush sizes instead of just the 3 colors it gave me. the undo feature was my own idea, I used getImageData and putImageData to save the canvas state after each stroke. the AI didnt mention

anything about undo. also the part where I convert the canvas to dataURL and send it as a message through firebase was all me.

2.3 GIF Picker (src/components/GifPicker.jsx, lines 1-57)

Prompt: "how to search gifs using giphy api in react"

Response: showed me the giphy search endpoint url and how to get the image url from the response json. pretty straightforward.

What I did: added debouncing with setTimeout (400ms) so it doesnt spam the api on every keystroke. also added trending gifs that load when you first open the picker, so its not just an empty box. the loading state and empty state handling was also me. the AI example was really bare bones, just the fetch call basically.

2.4 Reply to Message (src/components/Message.jsx, lines 101-125)

Prompt: "how to do reply to message in react chat app? like click reply and it shows the original msg, and clicking it scrolls back to that message"

Response: suggested using refs with scrollIntoView and storing replyTo info in the firestore message document (messageId, text, senderName).

What I did: the scrolling part I implemented myself using a refs map in ChatRoom.jsx. I also added a highlight animation in CSS so when you click a reply it flashes the original message yellow for a bit so you can actually see it. also handled the case where someone unsent the original message - it shows 'This message was deleted' instead of crashing. the AI didnt think about that edge case at all, I found it when I was testing.

2.5 Block User (src/components/BlockUserModal.jsx, lines 49-63)

Prompt: "how to do the blocking user in firestore? I want to store it as a list and filter out the message in the chat"

Response: said to use arrayUnion and arrayRemove on a blockedUsers field in the user document. showed a basic block/unblock function.

What I did: the arrayUnion/arrayRemove part is pretty standard so thats similar to what it gave. but the search UI, the list of currently blocked users, and the whole modal design was all me. also the part in ChatRoom.jsx where I actually filter out messages from blocked users in group chats, that logic was mine. the AI only showed the firestore part, not how to wire it into the actual chat.

2.6 Chrome Notifications (src/pages/ChatPage.jsx)

Prompt: "how to do browser push notification when new message comes in react"

Response: showed `Notification.requestPermission()` and new `Notification()`. very basic.

What I did: added the `document.hasFocus()` check so it only shows notification when you're not on the tab. the tricky part was making it work with the firestore `onSnapshot` listener so it only notifies for new messages and not all the old ones that load when you first open the chat. that took a while to figure out.

2.7 General CSS stuff

asked chatgpt a few times about CSS stuff like how to make sidebar stay fixed while the main area scrolls, and how to center a modal overlay. these are pretty common things and I just wrote the CSS myself after understanding the concept. didn't copy any CSS from it.

3. Summary

I mostly used AI to figure out how certain APIs work (gemini, giphy, canvas, browser notification). for each part I read what it gave me, understood it, then wrote my own version with extra stuff added. the overall project structure, firebase data model, components, and state management I came up with myself. but like I said I did use AI to polish and clean up a lot of my code after writing it, like fixing variable names, cleaning up messy logic, removing redundant code, stuff like that. so it's hard to say exactly which lines are 100% mine vs AI-touched, but the core ideas and logic are mine.